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The role of metabolism in the pathogenesis of systemic sclerosis

Honglin Zhu, Weilin Chen, Di Liu, Hui Luo *

Department of Rheumatology, Xiangya Hospital, Central South University, 87 Xiangya Road, Changsha, Hunan 410008, People's Republic of China

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ABSTRACT

Systemic sclerosis (SSc) is an immune-mediated autoimmune disease characterized by fibrosis and vascular abnormalities. The cellular and molecular mechanisms remain unclear, and current therapies are limited. Cell metabolism has been shown to play an essential role in cancer survival and tumour invasion as well as in rheumatic diseases such as systemic lupus erythematosus, rheumatoid arthritis and osteoarthritis. Although little is known about SSc, cell metabolism may provide new clues for understanding its pathogenesis. In this review, we summarize recent studies of metabolism in SSc and fibrotic disease, specifically focusing on glycolysis, fatty acid metabolism and oxidative stress. We highlight the role of metabolism in fibroblast differentiation and emphasize its potential therapeutic prospects in SSc.

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1. Introduction

Systemic sclerosis (scleroderma, SSc) is an immune-mediated, highly heterogeneous rheumatic disease characterized by fibrosis of the skin and internal organs and vascular abnormalities including Raynaud's phenomenon (RP) and pulmonary artery hypertension [1,2]. The prevalence of SSc is estimated at ~1/10,000 in the general population; however, mortality is highest in those with rheumatic disease [1]. The main cause of death is usually lung fibrosis or pulmonary arterial hypertension. SSc is divided into two clinical subsets based on the extent and severity of skin and organ involvement: diffuse cutaneous SSc (dSSc) and limited cutaneous SSc (lSSc). In dSSc, fibrosis is widespread throughout the skin as well as the visceral organs, while in lSSc, fibrosis is primarily restricted to the hands, arms and face [3]. The mechanism of SSc is very complicated. Self-perpetuating autoimmunity,

* Corresponding author.
E-mail address: luohui@csu.edu.cn (H. Luo).

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which activates both innate and adaptive immunity, plays a critical role in disease progression and leads to the activation and differentiation of mesenchymal cells and the accumulation of extracellular matrix. Immune cells, platelets, endothelial cells, fibroblasts and many autocrine and paracrine factors all contribute to its pathogenesis [4,5]. Great efforts have been made to elucidate the mechanism of SSc in recent years; however, much remains unknown. Effective treatments are not currently available.

Research into cellular metabolism and immunology has been emerging in recent decades [6–9]. There are six major metabolic pathways: glycolysis, the tricarboxylic acid (TCA) cycle, the pentose phosphate pathway (PPP), fatty acid oxidation, fatty acid synthesis and amino acid metabolism. Evidence has suggested that tumour cells convert glucose into lactic acid in the presence of oxygen, which is essential for cell survival and invasion (aerobic glycolysis or the Warburg effect) [10]. In addition to tumour cells, many other cell types, such as T cells, macrophages, dendritic cells, myeloid-derived suppressor cells and fibroblasts, also affect metabolic pathways depending on various environmental factors. Different immune cells use distinct metabolic signatures tailored to each biological function. Metabolic reprogramming plays a key role in regulating immune cell function and inflammatory responses and has been studied in rheumatologic diseases [11–16]. In systemic lupus erythematosus, mitochondrial glucose oxidation and hyperpolarization are increased in activated T cells [17]. Normalization of CD4+ T cell metabolism can reverse lupus in a mouse model [18]. In rheumatoid arthritis, decreased glycolysis, glucose shunting into the PPP [19,20] and low ATP levels [21] were observed in naive CD4+ T cells. In fibrotic disease, glycolysis was increased after fibroblast activation [14,15]. However, little is known about cellular metabolism in cases of SSc. Here, we summarize the current understanding of the metabolic aspects of SSc, highlight emerging concepts with potential value for therapy, and discuss the remaining problems to be resolved. This review summarizes new concepts related to metabolism in SSc and new insights for disease therapy in the clinical setting.

2. Altered Metabolism in SSc

Since the 2000s, positron emission tomography (PET) using the glucose analogue tracer ^{18}fluorodeoxyglucose (^{18}F-FDG) has been utilized extensively to detect tumours in the clinical setting. ^{18}F-FDG is dependent on the rate of glycolysis and can be used to quantify glucose uptake [22]. In SSc, PET/CT scans showed increased ^{18}F-FDG uptake in the skin and soft tissue calcinosis [23,24]. SSc patients with interstitial lung disease also demonstrated enhanced ^{18}F-FDG accumulation in the region around the pulmonary lesion, which was positively correlated with disease activity [25]. These results indicated that SSc patients exhibit increased glucose uptake and metabolic reprogramming.

3. Glycolysis

To date, no study has directly investigated the role of glycolysis in SSc. However, research about the role that glycolysis plays in the differentiation of fibroblasts and in fibrotic diseases is rapidly emerging. In the lung tissue of patients with idiopathic pulmonary fibrosis (IPF), mass spectrometry has suggested disruption of amino acid metabolism, glycolysis, the TCA cycle, and mitochondrial beta-oxidation [14,26]. Augmented glycolysis was found in lung fibroblasts isolated from IPF patients and in TGF-β1-treated normal lung fibroblasts. Glycolytic enzymes, including PFKFB3, PFK1 and HK2, were upregulated, and inhibition of PFKFB3 could potentially blunt the differentiation of fibroblasts and attenuate the profibrotic phenotype. Inhibition of glycolysis can decrease the extent of fibrosis in vivo [27]. In bleomycin-induced lung fibrosis mice, increased glycolysis played a critical role in fibroblast differentiation and fibrotic progression, and GLUT1-activated and AMP-activated protein kinase and hypoxia-inducible factor (HIF) 1α/PDK1 participated in fibrotic progression. Inhibition of GLUT1, HIF-1A or PDK1 resulted in a significant reduction in fibrosis [15,28,29]. Increased glycolysis and fatty acid oxidation were also found in the profibrotic M2-like profile of fibrotic lung alveolar macrophages isolated from bleomycin- or active TGF-β1-induced fibrotic mouse lungs [30]. In biopsy samples taken from patients with cirrhosis, carbon tetrachloride (CCl_4) or bile duct ligation induced hepatic fibrosis in mice, and the number of glycolytic stromal cells was positively correlated with the severity of fibrosis. High levels of glycolytic enzymes, such as HK2, PKM2, PFKP, GLUT1, and MCT4, the lipogenic gene peroxisome proliferator-activated receptor γ (PPARγ), and lactate, were present during the transdifferentiation of cultured, quiescent hepatic stellate cells (HSCs) into myofibroblasts. This process is mediated by HIF1α and Hedgehog signalling, which control the fate of HSCs via metabolic regulation. Inhibition of glycolysis, lactate accumulation, Hedgehog signalling or HIF1α can reverse myofibroblasts into quiescent HSCs [31,32]; lactate can also induce collagen expression in baboon liver fibroblasts [33]. High levels of glucose uptake, lactate production and glycolytic enzymes (HK, PKM2, PDK1, and PDK4) were found in unilateral ureter obstruction (UUO)-induced renal fibrosis mice and in TGFβ1-treated renal interstitial fibroblasts. Increasing glycolysis significantly induced the activation of myofibroblasts. Inhibiting glycolysis via 2-deoxyglucose (2DG) and shikonin (reducing PKM2) can reduce mouse renal fibrosis [34]. Taken together, the currently available evidence indicates that glycolysis is critical in fibroblast differentiation and the progression of fibrosis. Multiple interventions can effectively reverse fibrosis both in vitro and in vivo, providing valuable directions for further studies of SSc.

4. Fatty Acid Metabolism

Early in the 1970s, intradermal adipose tissue was found to be atrophied and replaced by collagen-rich fibrous tissue in SSc [35]. Adipocyte progenitor cells, which are essential for limiting fibrosis, are lost during the process of skin fibrosis [36]. Enhanced fibrosis with reduced body fat mass accumulation and small-sized adipocytes were found in lipodystrophy patients and TGF-β1 transgenic fibrosis mice [37,38]. Recently, dermal white adipose tissue (dWAT) was identified as a unique adipose tissue that is distinct from hypodermal adipose tissue and may play a critical role in skin barrier function. dWAT was reduced in SSc patients and fibrosis mice, and this reduction adversely affected skin homeostasis and promoted dermal fibrosis [39]. Improved dWAT loss, dermal fibrosis and survival were observed when fibrosis model mice were injected intradermally with adipose-derived stem cells [36,40]. In addition to dWAT loss, many of the canonical adipogenic markers, including fatty acid-binding protein 4 (FABP4), PPARγ2 and adiponectin, were also reduced in skin lesions. Adiponectin-positive intradermal progenitors differentiated into dermal myofibroblasts [41]. PPARγ is critical for adipocyte differentiation and has proven to be a master regulator of adipogenesis. Pharmacological PPARγ activation in bleomycin-induced skin fibrosis mice promotes systemic adipogenesis and attenuates inflammation and dermal fibrosis [42]. The Wnt/β-catenin pathway, which is critical for fibroblast activation, also disrupts adipocyte differentiation and induces fibrosis [43]. All these studies suggest an important connection between fibrosis and adipose tissue. Thus, fatty acid metabolism might be a contributing factor.

Fatty acids are essential for maintaining cell membrane integrity, signal transduction and energy production and regulating inflammation [44–46]. The monounsaturated free fatty acid oleate can increase the secretion of TGF-β, collagen I, fibronectin, and alpha smooth muscle actin (α-SMA) and induce a myofibroblast phenotype in mesangial cells [47]. In the plasma and red cell membranes of SSc patients, the levels of the prostaglandin precursors dihomo-gamma-linolenic acid (DGLA) and arachidonic acid (AA) and metabolites of AA (22:4n-6, 22:5n-6), which have important roles in the maintenance of normal cell membrane characteristics, were found to be exceptionally low. Supplementation with essential fatty acids has been found to be beneficial in SSc [48].

Considerable evidence has suggested that fatty acid metabolism is critical in fibrosis and other fibrotic diseases. The inflammatory and metabolic pathways in tubulointerstitial fibrosis patients were significantly altered. Critical enzymes in the regulation of fatty acid oxidation were decreased in tubulointerstitial fibrosis patients and mice, and increased intracellular lipid deposition was also noted. Inhibition of fatty acid oxidation in tubule epithelial cells led to ATP depletion, cell death, dedifferentiation and intracellular lipid deposition, similar to the effects observed in fibrosis. Using genetic or pharmacological methods to restore fatty acid metabolism can protect mice from tubulointerstitial fibrosis [49]. In cystic fibrosis, the release of AA from cell membrane phospholipids was increased, and lower levels of linoleic acid (LA) and docosahexaenoic acid (DHA) and disturbances in annexins and ceramides were described [50]. In porcine aortic valvular interstitial cells, exogenous polyunsaturated fatty acids (PUFAs), DHA and AA were able to reverse the myofibroblastic phenotype mediated by RhoA/G-actin/MRTF signalling [51]. In human prostate carcinoma, DHA inhibited TGF-β-induced myofibroblast differentiation and matrix metalloproteinase-2 (MMP2) production and suppressed the epithelial-mesenchymal transition (EMT) [52]. In cardiac fibroblasts, omega-3 PUFAs [eicosapentaenoic acid (EPA) and DHA] from fish oil blocked TGF-β1-induced phospho-Smad2/3 nuclear translocation through activation of the cyclic GMP/protein kinase G pathway and prevented cardiac fibrosis and dysfunction [53]. Nitrate-derived unsaturated fatty acids display pleiotropic behaviours and are critical in the inhibition of inflammatory cell function [54]. Nitrated fatty acids (NFAs) upregulated PPARγ, converted TGF-β to inactive monomers in a cell-free solution and blocked TGF-β signalling and activity in human lung fibroblasts. NFAs reduced disease severity, reversed myofibroblasts and decreased collagen deposition in an experimental pulmonary fibrosis mouse model [55]. Omega-3 PUFA derivatives, resolvins, have an anti-inflammatory effect by reducing neutrophil infiltration, paw oedema and proinflammatory cytokine expression [56]. Resolvin inhibited PDGF-BB-induced fibroblast proliferation and induced transient activation of the pro-proliferative ERK and AKT signalling pathways; it also exerted direct anti-fibrotic effects on UUO mice [57].

5. TCA Cycle

Intermediate metabolites in the TCA cycle were studied in fibrotic diseases. Succinate, which is formed in the TCA cycle, bound specifically to G protein-coupled receptor 91 (GPR91) and raised the levels of GPR91, α-SMA, TGF-β and type I collagen in HSCs. C57BL6/J mice fed an MCD diet (inhibitor of succinate dehydrogenase, SDH) exhibited elevated plasma succinate levels, increased succinate concentrations, and GPR91 and alpha-SMA expression in isolated HSCs [58]. Succinate was significantly upregulated in lung myofibroblasts and human fibrotic lungs with IPF, which markedly increased TGF-β1-induced HIF-1a expression and promoted fibroblast differentiation [27].

6. Amino Acid Metabolism

The lung consists of active tissue with rapid-turnover metabolism of both collagen and noncollagen proteins. In bleomycin-induced pulmonary fibrosis rabbits, an increased rate of synthesis, a decreased rate of degradation and rapid accumulation of collagen were found in the early stages of pulmonary fibrosis [59]. Genome-wide transcriptome analyses of kidney samples from severe tubulointerstitial fibrosis patients and mice suggested that carbohydrate, amino acid and lipid metabolism were markedly dysregulated [49]. Intracellular glutamine is utilized in the production of biological energy; synthesis of proteins, lipids and nucleotides; and maintenance of redox homeostasis. Glutamine 1 was upregulated in bleomycin-induced lung fibrosis and areas of human IPF. Aberrant glutamine metabolism plays a critical role in lung fibroblast differentiation [60]. Inhibition of glutamine metabolism attenuated bleomycin-induced lung injury, improved survival and diminished fibrosis by reducing Th17 cells and M2 macrophages [61]. An intimate connection between glutamine and Wnt signalling has been found, and modulated glutamine metabolism might provide a novel therapeutic method for SSc fibrosis [62] (Fig. 1).

7. Metabolic Reprogramming in Immune Cells

Each immune cell undergoes specific metabolic reprogramming during activation and differentiation, and these changes are essential for immunological functions. SSc is driven by the activation and effector functions of both innate and adaptive cells. Therefore, metabolic reprogramming in immune cells might play a central role in SSc pathogenesis [63,64].

7.1. T Cells

Activated T cells were found in the peripheral blood and lesional skin tissues of SSc patients, as indicated by increased T cell-derived cytokines in the serum and T cell activation markers in dermal tissues [65]. For CD4+ T cells, the balance of type 1 T helper (Th1)/Th2 cells and regulatory T cells (Tregs)/Th17 cells was disrupted in SSc patients [66,67]. Th1 cells can secrete interferon (IFN)-γ and inhibit collagen production and are decreased in SSc, whereas the Th2 cells are increased and can produce IL-4, IL-5, and IL-13 and promote mononuclear cell infiltration and fibrosis [68]. Increased levels of Th17 cells resulted in the activation of fibroblasts, vascular endothelial cells and smooth muscle cells through IL-17 [69]. Although the percentage of circulating Tregs in SSc patients is controversial, reduced functional capacity has been widely identified. Tregs secrete TGF-β and IL-10, suppress effector T cells and antigen-presenting cell activation, and contribute to the pathogenesis of SSc [70]. Th9 and Th22 cells, which produce IL-9 and IL-22, respectively, were increased in the circulation of SSc patients and were found to be associated with disease activity or complications [71,72]. The role of T follicular helper (Tfh) cells, which secrete IL-21, has not been identified in SSc and needs to be further confirmed. Among CD8 + T cells, the proportions of effector (CD8 + CD45RA + CD27-) and effector/memory (CD8 + CD45RA-CD27-) cells in the peripheral blood are increased. The latter produce IL-13 and IFN-γ and infiltrate the lesional skin in the early stage of dSSc [73]. CD8+ T cells exert proinflammatory and profibrotic activities and induce cytotoxic tissue damage in SSc [74].

Metabolic reprogramming is closely correlated with T cell survival, development, activation, function and differentiation [75]. Naive T cells utilize small amounts of glucose to generate ATP and maintain homeostatic proliferation and survival. The TCA cycle and oxidative phosphorylation are their predominant metabolic pathways [76]. Upon activation, CD4+ T cells differentiate into diverse subsets with distinct metabolic requirements. All subsets rely on glycolysis and oxidative phosphorylation, and Th2 and Th17 cells exhibit even higher levels of glucose uptake [77]. Tregs utilize fatty acid oxidation, while Th17 cells rely on de novo fatty acid synthesis; the balance of fatty acid oxidation and synthesis contributes to the differentiation of Th17 cells and Tregs [78,79]. Upon activation, CD8+ T cells exhibit increases in glycolytic rate and oxidative phosphorylation and differentiate into effector cells. Memory T cells exhibit a decreased glycolytic rate, utilize fatty acid oxidation and have more spare respiratory capacity than naive T cells and thus have a rapid recall capacity upon challenge [75,80].

7.2. B Cells

Hyperactivated B cells were found in SSc patients, as indicated by the overexpression of autoantibodies. B cells drive proinflammatory/profibrotic cytokines, upregulate stimulatory CD19 receptors and impair inhibitory CD22 receptors. The percentage of naive B cells was increased, the level of memory B cells was decreased but the cells themselves markedly activated, and IL-10-producing B regulatory cells

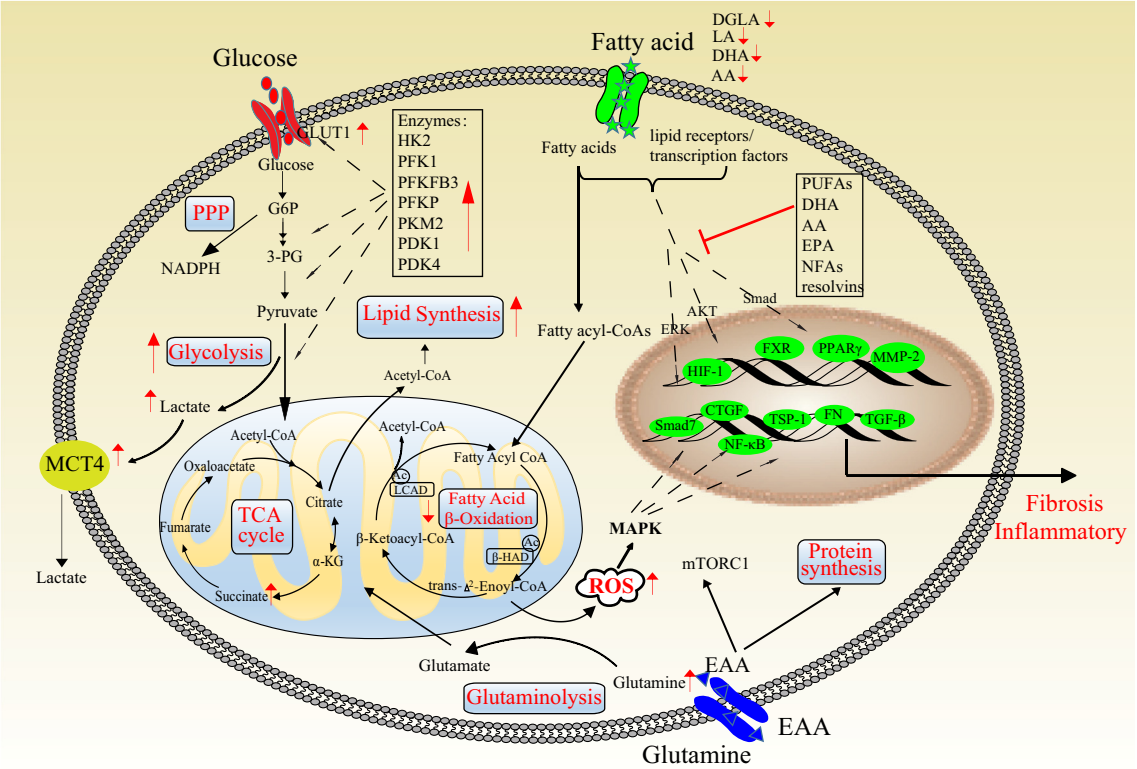

Fig. 1. Dysregulated metabolic pathways in the fibroblast of SSc or fibrotic diseases. Increased glycolysis, TCA cycle, and lipid synthesis and decreased fatty acid β-oxidation are found during fibroblast activation. Multiple essential fatty acid or derivatives (PUFAs, DHA, AA, EPA, NFAs, resolvins) can inhibit fibroblasts proliferation and transdifferentiation. Metabolic reprogramming results in redox imbalance and ROS accumulation, the latter activates multiple signalling pathways and exerts pro-fibrosis and pro-inflammatory effects. HK2, hexokinase 2; GLUT1, glucose transporter 1; PFK-1, phosphofructokinase-1; PFKFB3, fructose-2,6-biphosphatase 3; PFKP, phosphofructokinase; PKM2, pyruvate kinase M 2; PDK1, pyruvate dehydrogenase kinase 1; PDK4, pyruvate dehydrogenase kinase 4; MCT4, monocarboxylate transporter 4; PPP, pentose phosphate pathway; G6P, glucose-6-phosphate; 3-PG, 3-phosphoglycerate; α-KG, α-ketoglutarate; DGLA, dihomo-gamma-linolenic acid; LA, linoleic acid; DHA, docosahexaenoic acid; AA, arachidonic acid; PUFAs, polyunsaturated fatty acids; EPA, eicosapentaenoic acid; NFAs, nitrated fatty acids; EAAs, essential amino acids; mTORC1, mTOR complex 1.

were reduced in SSc patients. Interestingly, the mechanistic target of rapamycin (mTOR) phosphorylation, which is a central regulator of immunity, was significantly decreased in B cells, suggesting a switch in metabolic pathways in B cells in SSc [81,82].

Different subsets of B cells have different metabolic needs. Upon activation, B cells rapidly increase glucose uptake [6]. High metabolic demands were found in the early proliferative stages of B cells in bone marrow, including the pre-B, immature, and transitional stages of the bone marrow and spleen, and even higher metabolic needs were found in plasma cells that secreted high-affinity antibodies. Energy needs in quiescent blood-borne naive and memory B cells were relatively low but increased during antigen-driven proliferation. The anabolic metabolism was dysregulated in activated B cells, glycolysis and glutaminolysis were increased, and oxidation was decreased. A high level of catabolic metabolism, including fatty acid oxidation and pyruvate oxidation, was found in memory B cells [83,84].

7.3. Other Immune Cells

Soluble CD163, a putative marker of type 2 (M2) macrophages, was increased in the peripheral blood of SSc patients and was correlated with a poor clinical outcome [85]. Abundant M2 macrophages infiltrated the lesional skin and lung tissues of SSc patients [86]. M2 macrophages play key roles in the process of tissue fibrosis. In activated lung-resident macrophages, the expression of lipid and cholesterol trafficking-related genes was increased, indicating metabolic reprogramming in these cells [87].

The frequency of plasmacytoid dendritic cells (pDC) was reduced in the peripheral circulation of SSc patients but increased in the affected skin tissues [86]. CXCL4 secretion by pDCs was significantly increased in the circulation and associated with SSc skin fibrosis and pulmonary arterial hypertension [88].

Elevated numbers of type 2 innate lymphoid cells (ILC2) were found in the peripheral blood and lesional skin of SSc patients, and the number of cells was correlated with the extent of cutaneous fibrosis [89]. Natural killer (NK) cells [90] and neutrophils [91] exhibited altered properties and phenotypes in the blood of SSc patients; however, their roles are unclear.

Overall, both innate and adaptive immune cells display obvious metabolic changes. The six metabolic pathways affect the activation and differentiation of immune cells and might play critical roles in SSc. Extensive research has been conducted to explore metabolic reprogramming under normal conditions. However, little is known about the metabolic changes in immune cells in the pathophysiology of SSc. Further investigations in this field will provide substantial information regarding the mechanism of SSc.

8. Oxidative Stress and SSc

Increasing evidence has indicated the presence of intimate crosstalk between redox balance and metabolism. Metabolic pathways, such as glycolysis, glutaminolysis, fatty acid oxidation, one-carbon metabolism and the PPP, can generate the antioxidant molecules NADPH and GSH as well as the redox cofactors NADH and FADH, which are essential for producing and removing reactive oxygen species (ROS) and maintaining redox homeostasis [92,93]. The role of oxidative stress in the pathogenesis of SSc has been widely studied in recent years. In early 1993, Murrel first proposed the important role of oxidative stress in the aetiology of SSc [94]. Subsequently, a series of studies supported this

hypothesis. In SSc, circulating oxidative stress markers, including nitric oxide, malondialdehyde (MDA), asymmetric dimethylarginine (ADMA) and hydroperoxides (ROOHs), were increased, and antioxidant components, such as superoxide dismutase and vitamin C, were decreased [95]. High levels of MDA were negatively correlated with disease duration. Carbonyl was inversely correlated with the modified Rodnan skin score (mRSS) and was decreased in patients with pulmonary fibrosis. ROOHs were positively correlated with the capillaroscopy semiquantitative rating scale score and the rating system for avascular areas [96,97]. The level of ADMA is a reflection of endothelial cell dysfunction and was shown to be a novel and important biomarker for the prediction of cardiovascular events [98]. Elevated serum 8-isoprostane was positively correlated with the severity of pulmonary fibrosis, extent of renal vascular damage and level of immunological abnormalities [99] and was also correlated with the heat shock protein 70 level; the latter was associated with pulmonary fibrosis, skin sclerosis, renal vascular damage, oxidative stress and inflammation [100]. Advanced oxidation protein products (AOPPs), which provide indirect evidence of oxidative stress, were upregulated, stimulating endothelial cells and fibroblasts to produce hydrogen peroxide (H_2O_2), and were involved in vascular or fibrotic complications [101]. Stimulatory PDGFR antibodies were elevated in all SSc patients and appeared to be a specific hallmark. These autoantibodies induced collagen gene expression and fibroblast activation via the Ha-Ras-ERK1/2 and ROS cascades [102]. In SSc skin biopsies, the oxidative stress-related proteins peroxiredoxin 1 and carbonyl reductase I were upregulated, as detected by proteomic analysis [103]. AOPPs were also increased in the skin of dSSc patients, as detected by noninvasive skin autofluorescence [104]. In urine samples, the oxidative production of free radical-catalysed peroxidation of AA, 8-oxo-2′-deoxyguanosine (8-oxodG) and isoprostanes were significantly increased. 8-OxodG was associated with the presence of pulmonary fibrosis and with decreased forced vital capacity and DLCO/alveolar volume. In dSSc, mRSS >14 was independently correlated with the 8-oxodG level [105,106]. Increased levels of 8-isoprostane and H_2O_2 were also found in SSc patients' exhalations [107,108]; however, no correlation with radiological findings was found.

An experimental SSc mouse model also revealed the important roles of oxidative stress. In the tight-skin (TSK-1/+) fibrosis mouse model, the expression levels of the protective antioxidant proteins endothelial nitric oxide synthase (eNOS), haemoxygenase-1 (HO-1) and nuclear factor erythroid 2-related factor 2 (Nrf2) in the skin were reduced, although no difference was found in the lung [109]. Overexpressed hydroxyl radicals and superoxide were found in bleomycin-induced mouse fibrosis [110]. More importantly, ROS-inducing agents induced widespread skin and lung fibrosis in BALB/c mice, directly suggesting that ROS play a key role in fibrosis [111]. Oxidative stress and antioxidant molecules are critical for fibroblast activation and function. Therefore, further studies are needed to investigate the major metabolic pathways controlling redox homeostasis in fibroblasts. Interventions targeting these metabolic pathways could facilitate the development of novel antifibrotic therapies (Table 1).

Table 1
Oxidative stress related molecules in SSc.

Sample	Molecules	Reference
SSc circulating	MDA↑	[95]
	Superoxide dismutase↓, vitamin C↓	[95]
	ADMA↑	[95,98]
	ROOHs↑	[95–97]
	8-Isoprostane↑	[99]
	AOPPs↑	[101]
	PDGFR antibodies↑	[103]
Skin biopsies	Peroxiredoxin 1↑, carbonyl reductase I ↑	[103]
	AOPPs ↑	[104]
Urine samples	8-oxodG↑	[105,106]
Exhalations	8-Isoprostane↑, H2O2↑	[107,108]
Skin tissues of tight-skin (TSK-1/+) fibrosis mouse	eNOS↓, HO-1 Nrf2↓	[109]
Bleomycin-induced mouse	Hydroxyl radicals↑, superoxide↑	[110]

9. Conclusion

Given the important roles of metabolism in the activation of immune cells and fibroblasts and in ROS production, the exploitation of metabolic reprogramming interventions may provide attractive and effective therapeutic approaches for SSc. Some targets have been used in clinical practice. The glucose analogue 2-DG and metformin suppressed glycolysis and mitochondrial metabolism. 2-DG plus metformin inhibited the production of cytokines and autoantibodies in lupus-prone B6.sle1.sle2.sle3 mice and normalized the metabolism of SLE CD4+ T cells [18]. PFKFB3 [112] and lactate dehydrogenase (LDH) [113] modulated glycolysis, reduced IFN-γ production and suppressed T cell activation. Targeting glycolysis could inhibit fibroblast differentiation and be beneficial for many fibrotic diseases, as previously mentioned. Methotrexate is an important immune system suppressant and is widely used in rheumatic diseases. Methotrexate can inhibit one-carbon metabolism, including amino acid metabolism and nucleotide synthesis, and suppress Janus kinase (JAK)-STAT signalling; it also plays important roles in cell growth, redox balance and epigenetics [114,115]. Due to the critical roles of fatty acid metabolism in SSc, targeting this pathway may be a promising treatment approach. Drugs that modify fatty acid metabolism or dietary supplementation with essential fatty acids could be beneficial in treating SSc.

Regulation of mitochondrial ROS production would also be an interesting target. The mitochondrial ROS scavenger MitoQ can reduce IFN-γ production [116,117]. Pyruvate dehydrogenase kinase 1 (PDHK1) regulates pyruvate flux into the TCA cycle and controls mitochondrial ROS production. Inhibition of PDHK1 promotes an increase in ROS, which induces Tregs and exerts a protective role in experimental autoimmune encephalitis [118].

Metabolic pathway analysis broadens our understanding of the mechanisms of SSc and provides insights into novel therapeutic agents. Each metabolic pathway has specific nutrient transporters, enzymes, and multiple metabolic steps, many of which can be used as targets to design pharmacological intervention.

Over the past decade, considerable evidence has suggested the critical role of metabolism in fibroblast differentiation, fibrotic diseases and SSc. Most studies have focused on glycolysis and fatty acid metabolism as well as oxidative stress and have provided new clues for exploring the mechanisms of SSc. However, immune or inflammatory cells, endothelial cells, and epithelial cells all play important roles in its pathogenesis. In addition to fibroblasts, studies about metabolic reprogramming in immune cells are valuable and urgently needed. Meanwhile, investigations into the TCA cycle, the PPP, and amino acid metabolism are almost non-existent and are urgently needed. Nevertheless, the currently available data have led us to propose that interventions in glycolysis, fatty acid metabolism or oxidative stress could benefit SSc fibrosis. Future research should focus on improving our understanding of metabolic mechanisms in immune cells, fibrosis and vascular abnormalities, on searching for available metabolic intermediates for disease biomarkers, and on identifying the most important enzymes for therapeutic targeting.

Accumulated data have indicated important contributions of metabolism to the pathogenesis of SSc and fibrotic diseases. Multiple approaches are available to restore the dysregulated metabolic pathways, such as 2-DG and metformin inhibit glycolysis, methotrexate suppresses one-carbon metabolism, and MitoQ decreases ROS production. Many of them exhibit anti-fibrosis effects in mouse model. Alterations in metabolite levels can affect gene expression and cell signalling. Some metabolites are correlated with SSc activity and the extent of fibrosis. Circulating metabolites have the potential to be novel diagnostic and prognostic biomarkers. However, more works are needed to explore metabolic targets and biomarkers for clinical use.

Author Contributions

HZ wrote first draft. WC and DL revised the manuscript. HL has revised final version and added extra information.

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Conflict of Interest

The authors declare that there is no conflict of interest regarding the publication of this paper.

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